Band-1 Hadamard order 11520 – Classical Paley-I (q=11519); IN SPK via Paley-I; Gap 2052 Still OPEN

H_11520 Paley-I PASS; Paley-II FAIL/SKIP (q=5759=13*443 NOT prime; ===3 mod 4 needs ===1); Kronecker FAIL/SKIP (4*2880; 2880===0 mod 4 order-admissible; not_in_seeds / no H_2880); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-30

Document class. Scientific computational / construction certificate (Band-1).

Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.

Parents. `2026-09-30_Band1_Hadamard_11492_PaleyI`; `2026-09-30_Band1_Hadamard_11484_PaleyI`; `2026-09-30_Band1_Hadamard_11472_PaleyI`; `2026-09-30_Band1_Hadamard_11468_PaleyI`; `2026-09-30_Band1_Hadamard_11448_PaleyI`; `2026-09-30_Band1_Hadamard_11444_PaleyI`; `2026-09-30_Band1_Hadamard_11424_PaleyI`; `2026-09-30_Band1_Hadamard_11412_PaleyI`; `2026-09-30_Band1_Hadamard_11400_PaleyI`; `2026-09-30_Band1_Hadamard_11384_PaleyI`; `2026-09-30_Band1_Hadamard_11352_PaleyI`; `2026-09-30_Band1_Hadamard_11312_PaleyI`; `2026-09-30_Band1_Hadamard_11300_PaleyI`; `2026-09-30_Band1_Hadamard_11288_PaleyI`; `2026-09-30_Band1_Hadamard_11280_PaleyI`; `2026-09-30_Band1_Hadamard_11252_PaleyI`; `2026-09-29_Band1_Hadamard_11244_PaleyI`; `2026-09-29_Band1_Hadamard_11240_PaleyI`; `2026-09-29_Band1_Hadamard_11172_PaleyI`; `2026-09-29_Band1_Hadamard_11160_PaleyI`; `2026-09-29_Band1_Hadamard_11132_PaleyI`; `2026-09-29_Band1_Hadamard_11120_PaleyI`; `2026-09-29_Band1_Hadamard_11088_PaleyI`; `2026-09-29_Band1_Hadamard_11084_PaleyI`; `2026-09-29_Band1_Hadamard_11072_PaleyI`; `2026-09-29_Band1_Hadamard_11060_PaleyI`; `2026-09-29_Band1_Hadamard_11048_PaleyI`; `2026-09-29_Band1_Hadamard_11028_PaleyI`; `2026-09-29_Band1_Hadamard_11004_PaleyI`; `2026-09-29_Band1_Hadamard_10988_PaleyI`; `2026-09-29_Band1_Hadamard_10980_PaleyI`; `2026-09-29_Band1_Hadamard_10940_PaleyI`; `2026-09-29_Band1_Hadamard_10892_PaleyI`; `2026-09-29_Band1_Hadamard_10884_PaleyI`; `2026-09-29_Band1_Hadamard_10860_PaleyI`; `2026-09-29_Band1_Hadamard_10848_PaleyI`; `2026-09-29_Band1_Hadamard_10832_PaleyI`; `2026-09-29_Band1_Hadamard_10800_PaleyI`; `2026-09-29_Band1_Hadamard_10772_PaleyI`; `2026-09-29_Band1_Hadamard_10740_PaleyI`; `2026-09-29_Band1_Hadamard_10724_PaleyI`; `2026-09-29_Band1_Hadamard_10712_PaleyI`; `2026-09-28_Band1_Hadamard_10692_PaleyI`; `2026-09-28_Band1_Hadamard_10688_PaleyI`; `2026-09-28_Band1_Hadamard_10668_PaleyI`; `2026-09-28_Band1_Hadamard_10652_PaleyI`; `2026-09-28_Band1_Hadamard_10640_PaleyI`; `2026-09-28_Band1_Hadamard_10632_PaleyI`;

2026-09-28_Band1_Hadamard_10628_PaleyI; 2026-09-28_Band1_Hadamard_10608_PaleyI;
2026-09-28_Band1_Hadamard_10568_PaleyI; 2026-09-28_Band1_Hadamard_10560_PaleyI;
2026-09-28_Band1_Hadamard_10532_PaleyI; 2026-09-28_Band1_Hadamard_10500_PaleyI;
2026-09-28_Band1_Hadamard_10488_PaleyI; 2026-09-28_Band1_Hadamard_10464_PaleyI;
2026-09-28_Band1_Hadamard_10460_PaleyI; 2026-09-28_Band1_Hadamard_10428_PaleyI;
2026-09-28_Band1_Hadamard_10400_PaleyI; 2026-09-24_Band1_Hadamard_10392_PaleyI;
2026-09-24_Band1_Hadamard_10344_PaleyI; 2026-09-24_Band1_Hadamard_10332_PaleyI;
2026-09-24_Band1_Hadamard_10304_PaleyI; 2026-09-24_Band1_Hadamard_10272_PaleyI;
2026-09-23_Band1_Hadamard_10268_PaleyI; 2026-09-23_Band1_Hadamard_10224_PaleyI;
2026-09-23_Band1_Hadamard_10212_PaleyI; 2026-09-23_Band1_Hadamard_10164_PaleyI;
2026-09-23_Band1_Hadamard_10160_PaleyI; 2026-09-23_Band1_Hadamard_10152_PaleyI;
2026-09-23_Band1_Hadamard_10140_PaleyI; 2026-09-23_Band1_Hadamard_10112_PaleyI;
2026-09-23_Band1_Hadamard_10104_PaleyI; 2026-09-23_Band1_Hadamard_10100_PaleyI;
2026-09-23_Band1_Hadamard_10068_PaleyI.
Lab. `/workspace/experiments/band1/hadamard_11520_paleyI/`.
Cutoff. 2026-09-30T20:50:04Z (~4:50PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite construction of order 11520 only (verified max|HHˆT-nI|=0), via classical Paley-I q=11519 (prime ===3 mod 4). 11520 IN SPK via Paley-I.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 11520: Paley-I **PASS** ($q = 11519$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL/SKIP** (q=5759=13*443 NOT prime; ===3 mod 4 needs ===1); Kronecker **FAIL/SKIP** (note $11520 = 4 \cdot 2880$; $2880 \equiv 0 \pmod 4$ order-admissible; not_in_seeds / no H_{2880} (factors $2^6 \cdot 3^2 \cdot 5$)). **11520** folds into SPK **only** via classical Paley-I (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **11520+11504+11492+11484+11472+11468+11448+11444+11424+11412+11400+11384+11352+113** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

ewpage

Abstract

Finite construction of order **11520**.

- H_{11520} = Paley-I ($q = 11519$) – classical SPK
- Paley-II for 11520: **FAIL/SKIP** ($q = 5759 = 13 \cdot 443$ NOT prime; $\equiv 3 \pmod 4$ needs $\equiv 1$)
- Kronecker for 11520: **FAIL/SKIP** ($4 \cdot 2880$; $2880 \equiv 0 \pmod 4$ order-admissible; not_in_seeds / no H_{2880} (factors $2^6 \cdot 3^2 \cdot 5$))
- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-I.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_11520.py`

Results. overall_pass=True; units 368/368; row blocks 90/90 (block size 128); wall 1237.0304s; constructed max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 11520 = q + 1$ with $q = 11519$ prime $\equiv 3 \pmod 4$ — Paley-I eligible.
- Paley-II: $q = 5759 = 13 \cdot 443$ NOT prime; $\equiv 3 \pmod 4$ needs $\equiv 1$ — FAIL/SKIP.
- Kronecker: $11520 = 4 \cdot 2880$; $2880 \equiv 0 \pmod 4$ order-admissible; not_in_seeds / no H_{2880} (factors $2^6 \cdot 3^2 \cdot 5$) — FAIL/SKIP.
- Miyamoto digression skipped this wave (prefer classical SPK).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_11520 Paley-I	Paley-I q=11519	0	1213.9826
H_11520 Paley-II	FAIL/SKIP (q=5759=13*443 NOT prime; ===3 mod 4 needs ===1)	–	–
H_11520 Kronecker	FAIL/SKIP (4*2880; 2880===0 mod 4 order-admissible; not_in_seeds / no H_2880)	–	–

SPK closure note

Seeds include prior SPK through **11504** lineage (**11504**; **11492**; **11484**; **11472**; **11468**; **11448**; **11444**; **11424**; **11412**; **11400**; **11384**; **11352**; **11312**; **11300**; **11288**; **11280**; **11252**; **11244**; **11240**; **11172**; **11160**; **11132**; **11120**; **11088**; **11084**; **11072**; **11060**; **11048**; **11028**; **11004**; **10988**; **10980**; **10940**; **10904**; **10892**; **10884**; **10860**; **10848**; **10832**; **10800**; **10772**; **10740**; **10724**; **10712**; **10692**; **10688**; **10668**; **10652**; **10640**; **10632**; **10628**; **10608**; **10568**; **10560**; **10532**; **10500**; **10488**; **10464**; **10460**; **10428**; **10400**; **10392**; **10344**; **10332**; **10304**; **10272**; **10268**; **10260**; **10248**; **10244**; **10224**; **10212**; **10164**; **10160**; **10152**; **10140**; **10112**;

10104; **10100**; **10068**; **10040**; **10008**; **9968**; **9932**; **9924**; **9908**; **9888**; **9884**; **9872**; **9860**; **9852**; **9840**; **9812**; **9804**; **9792**; **9788**; **9768**; **9744**; **9740**; **9720**; **9680**; **9644**; **9632**; **9624**; **9620**; **9588**; **9552**; **9548**; **9540**; **9512**; **9492**; **9480**; **9468**; **9420**; **9404**; **9392**; **9372**; **9344**; **9324**; **9320**; **9312**; **9284**; **9240**; **9228**; **9204**; **9200**; **9188**; **9152**; **9104**; **9128**; **9124**; **9092**; **2220+8884**; **2236**; **2160**; **2140+8564**; . . . ; **NOT** 8948/8644/8516/. . .) + verified **10464** (+ prior **10392+10344+10332+10304+10272+10268+10224+10212+10164+10160+10152+10140+10112+101**
Next constructive gap **2052** (OPEN).

Quantity	Value
11520 in SPK	**True** (Paley-I q=11519)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

Next optional

q=11527->n=11528 Paley-I DIRECT q=11527 OR gap 2052 / Track B OR n=2052 standard SPK constructive gap – method-attack / lit OR Track B / Object 1 meter await